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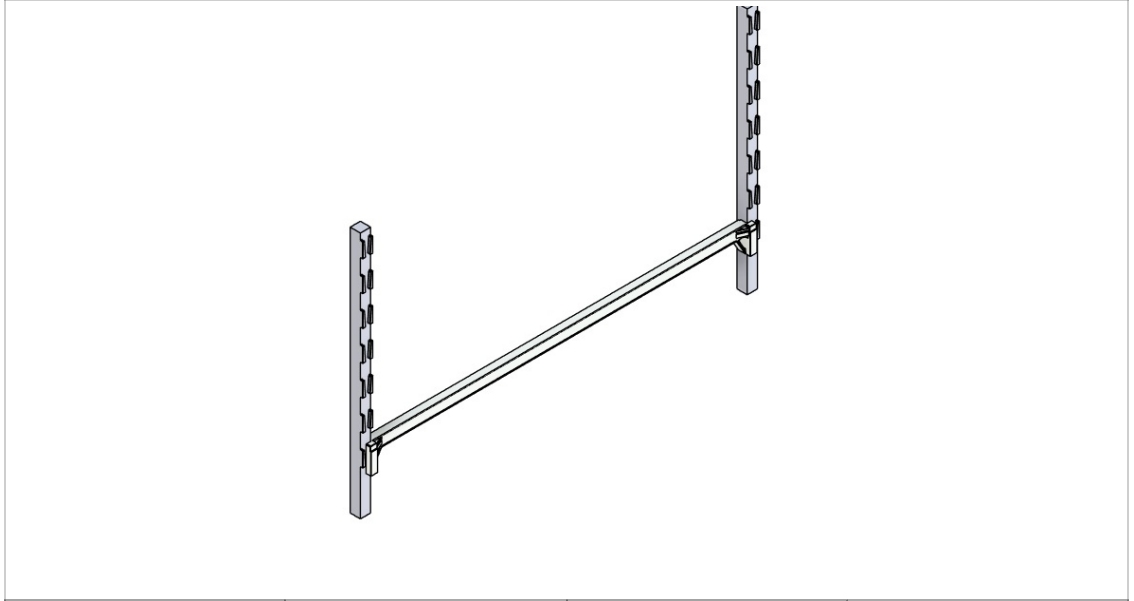
Result Summary

Safety Factor

Stress

Displacement

Study Report



Analyzed File	3rd Simulation v4	Version	Autodesk Fusion (2604.1.48)
Creation Date	2025-10-24, 08:37:41	Author	User
Summary	Connected some shit. but the shelf shit is steel. need more data		

Report Properties

Report Properties	
Title	Studies
Author	User

Simulation Model 1

Study 1 - Static Stress

Study Properties

Study Properties	
Study Type	Static Stress
Last Modification Date	2025-10-24, 08:31:35

Study Setup

Settings		
General		
	Contact Tolerance	0.10 mm
	Remove Rigid Body Modes	No
Mesh		
	Average Element Size (% of model size) - Solids	10
	Scale Mesh Size Per Part	Yes
	Average Element Size (absolute value)	-
	Element Order	Linear
	Create Curved Mesh Elements	No
	Max. Turn Angle on Curves (Deg.)	60
	Max. Adjacent Mesh Size Ratio	1.5
	Max. Aspect Ratio	10

	Minimum Element Size (% of average size)	20
Adaptive Mesh Refinement		
	Number of Refinement Steps	0
	Results Convergence Tolerance (%)	20
	Portion of Elements to Refine (%)	10
	Results for Baseline Accuracy	von Mises Stress

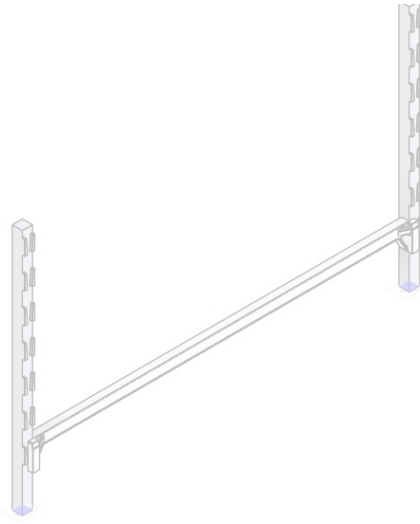
Materials		
Steel		
Components		
	Component	Strength
	Side_Shel lock Final v1:1/Other Side Shelf Lock v1:1	Yield Strength
	Smalle-Leg-Finalr v1 (1):1	Yield Strength
	Smalle-Leg-Finalr v1:1	Yield Strength
Properties		
	Density	7.850E-06 kg / mm^3
	Young's Modulus	210000.00 MPa
	Poisson's Ratio	0.30
	Yield Strength	207.00 MPa
	Ultimate Tensile Strength	345.00 MPa
	Thermal Conductivity	0.056 W / (mm C)
	Thermal Expansion Coefficient	1.200E-05 / C
	Specific Heat	480.00 J / (kg C)

Contacts				
Contact Type	Contact Set	Penetration Type	Bodies	Entities
Bonded				
	Bonded3	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1 (1):1]	[Face 1019 Face 1219]
	Bonded6	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1:1]	[Face 1004 Face 1228]
	[M] Bonded8	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1:1]	[Face 1009 Face 1230]
	[M] Bonded9	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1 (1):1]	[Face 1030 Face 1217]
	Bonded10	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1 (1):1]	[Face 1133 Face 1408]
	Bonded11	Symmetric	[Body1 Other Side Shelf Lock v1:1 Body1 Smalle-Leg-Finalr v1:1]	[Face 898 Face 1406]

Mesh	
Type	Solids
Nodes	50923
Elements	240410

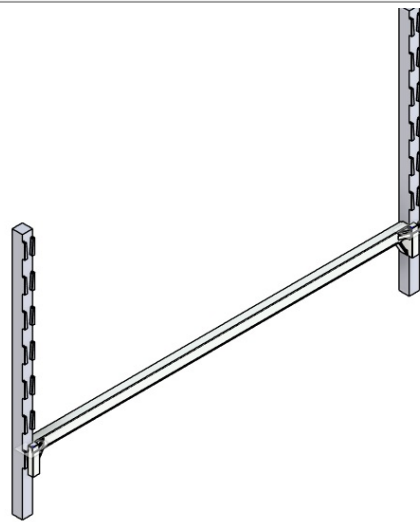
Load Case1

Constraints
Fixed1



Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

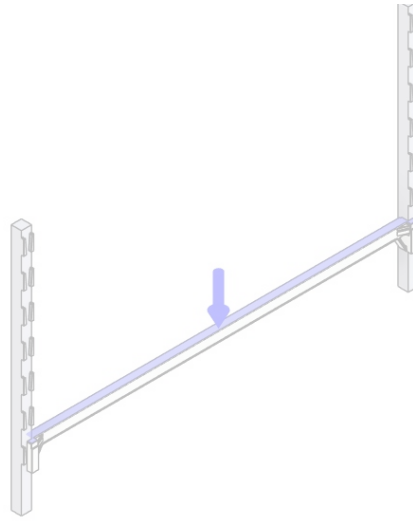
Fixed2



Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

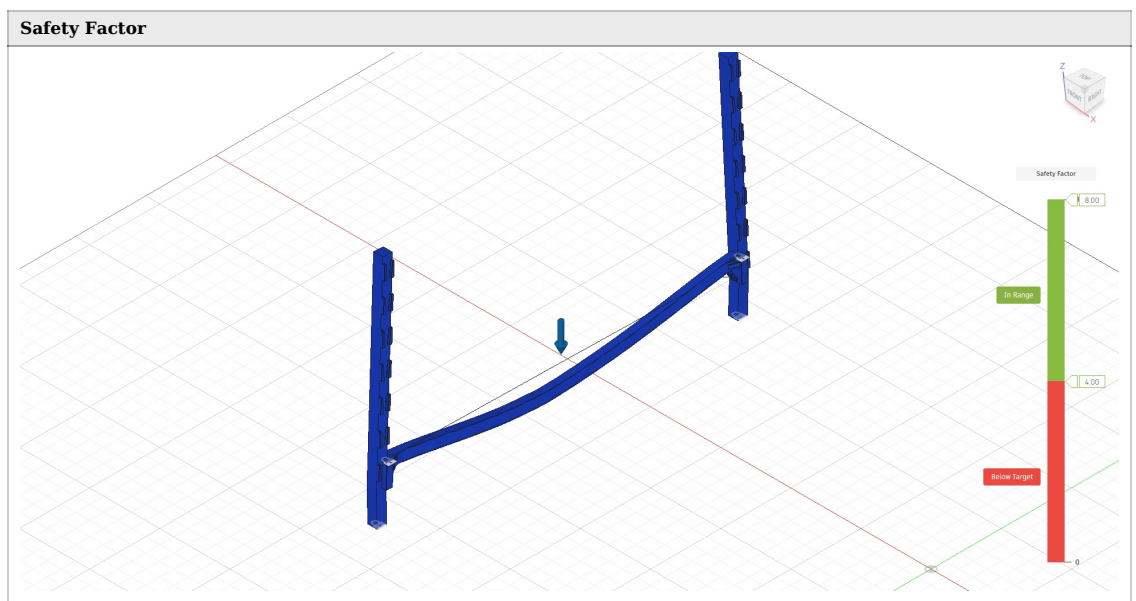
Loads

Force1



Type	Force
Magnitude	4070.123 N
X Value	0.00 N
Y Value	0.00 N
Z Value	-4070.123 N
Force Per Entity	No

Load Case1 - Guided Results



● With the analysis criteria in this study setup, the design is expected to bend permanently or break.

Below Safety Factor Target

Validate your study setup

- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results

- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Strengthen weak areas

- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Generate alternative designs

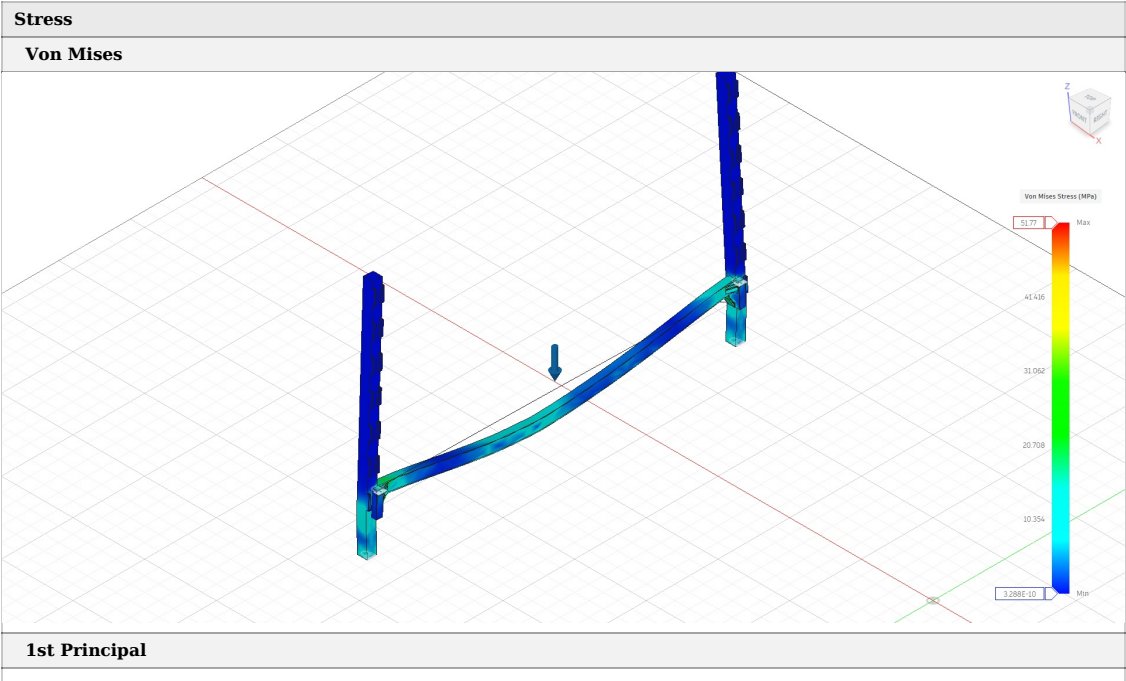
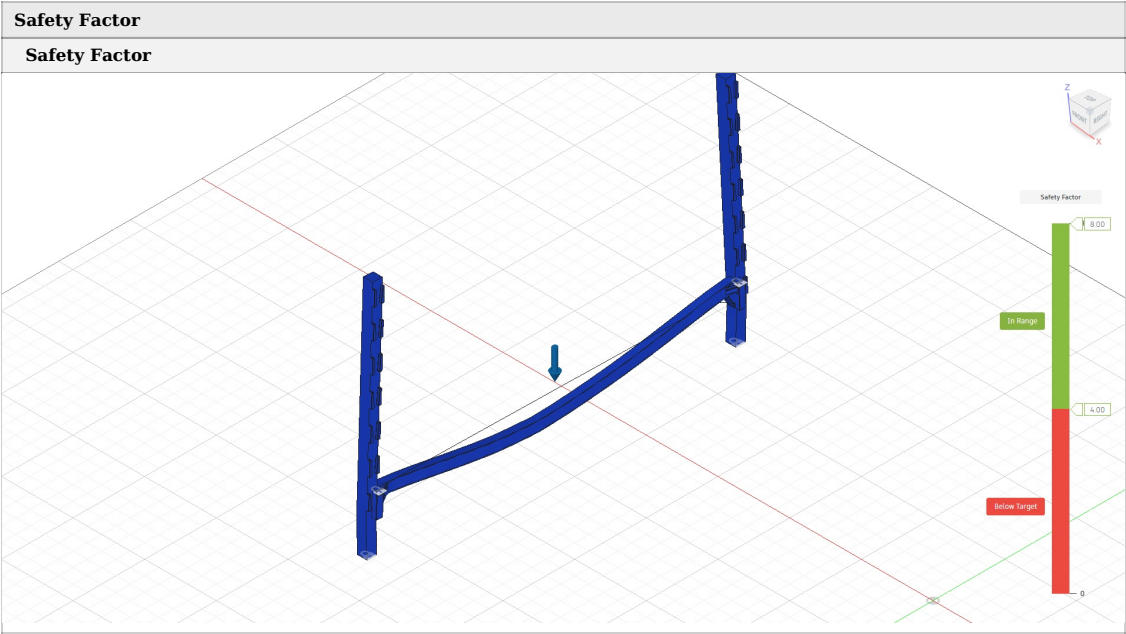
- Add more material to weak areas

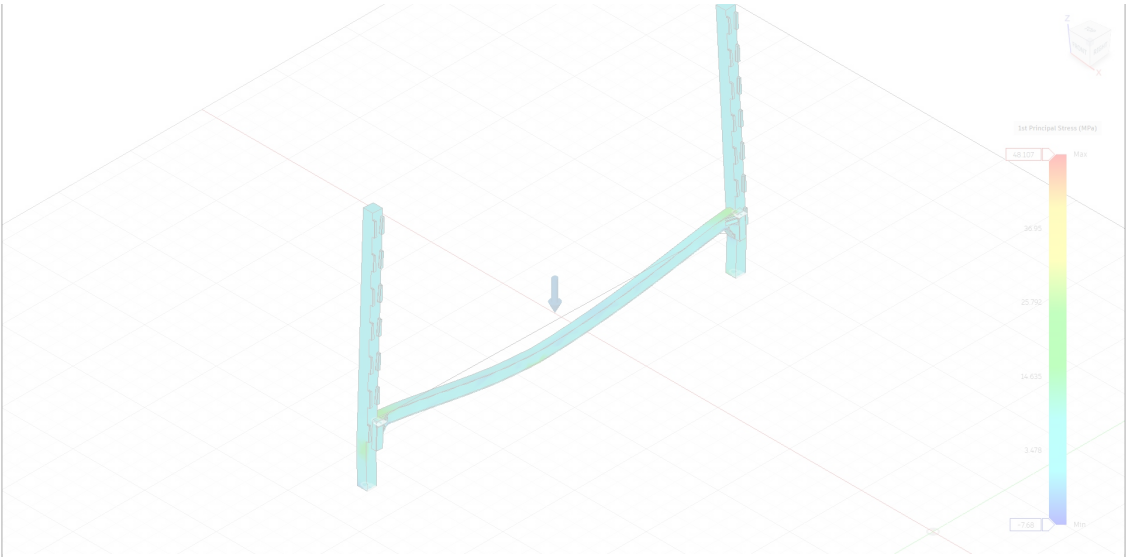
Next Steps	• Select a material with a higher yield strength • Stiffen the design • Modify the shape with a Generative Design study
	Meets Safety Factor Target
	Validate your study setup • Check the loads are not set too high • Check the mesh refinement and consider increasing it in areas of high stress • Confirm your Safety Factor requirement • If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight
	Check other results • Examine the Below target Result visualization to see where the weakest areas are • Check the stress result to see if, and where, the material might yield • Check the displacement result to see how far from its original alignment the model has moved • Look at other results, such as reaction forces to check your study set up, or strain to check your material performance
	Above Safety Factor Limit
	Validate your study setup • Check the loads are not set too high • Check the mesh refinement and consider increasing it in areas of high stress • Confirm your Safety Factor requirement • If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight
	Check other results • Examine the Below target Result visualization to see where the weakest areas are • Check the stress result to see if, and where, the material might yield • Check the displacement result to see how far from its original alignment the model has moved • Look at other results, such as reaction forces to check your study set up, or strain to check your material performance
	Lightweight the design • Set a Safety Factor upper limit to find where material can be removed • Remove material from areas that exceed the Safety Factor upper limit
	Generate alternative designs • In the Generative Design workspace, create design modifications • Remove unnecessary material using a Shape Optimization study

Load Case1 - Results

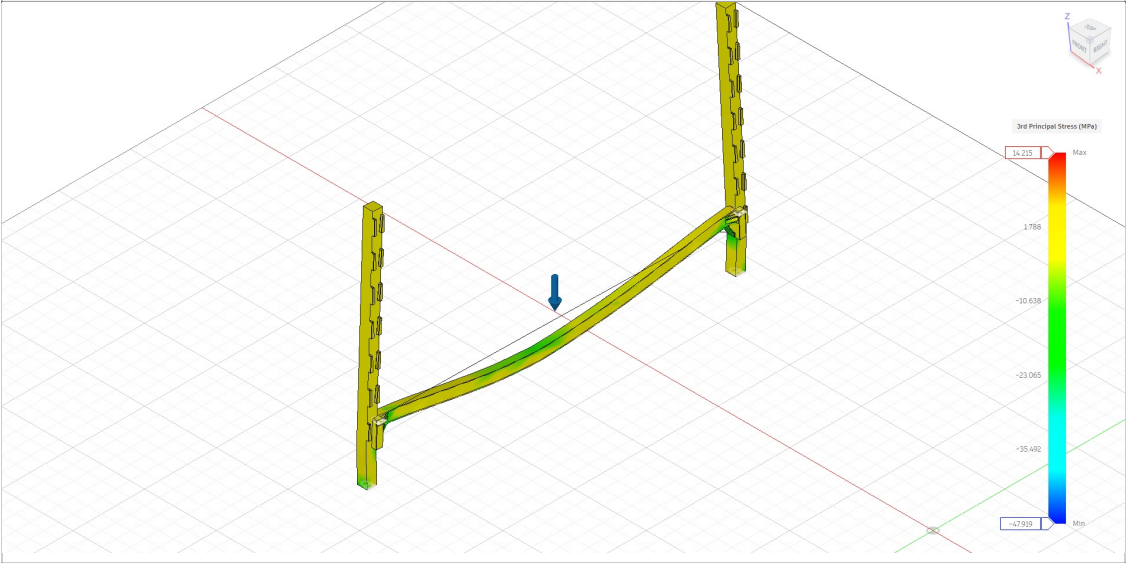
Result Summary			
Result type	Sub-result	Minimum	Maximum
Safety Factor			
	Safety Factor	3.998	6.295E+11
Stress			
	Von Mises	3.288E-10 MPa	51.77 MPa
	1st Principal	−7.68 MPa	48.107 MPa
	3rd Principal	−47.919 MPa	14.215 MPa
	Normal XX	−13.315 MPa	15.925 MPa
	Normal YY	−20.817 MPa	44.332 MPa
	Normal ZZ	−27.60 MPa	24.476 MPa
	Shear XY	−20.277 MPa	16.608 MPa
	Shear YZ	−21.498 MPa	23.202 MPa
	Shear XZ	−9.878 MPa	16.296 MPa
Displacement			
	Total	0 mm	0.282 mm
	X	−0.03 mm	0.023 mm
	Y	−0.103 mm	0.105 mm
	Z	−0.281 mm	0.005 mm
Reaction Force			
	Total	0 N	3,030.124 N
	X	−610.941 N	827.464 N
	Y	−2,153.644 N	2,030.29 N
	Z	−2,286.929 N	2,393.297 N
Strain			

	Equivalent	1.798E-15	4.153E-04
	1st Principal	−6.729E-12	3.543E-04
	3rd Principal	−4.135E-04	2.385E-11
	Normal XX	−3.732E-05	8.001E-05
	Normal YY	−8.263E-05	1.84E-04
	Normal ZZ	−1.187E-04	9.573E-05
	Shear XY	−2.511E-04	2.056E-04
	Shear YZ	−2.662E-04	2.873E-04
	Shear XZ	−1.223E-04	2.018E-04
Contact Pressure			
	Total	0 MPa	44.08 MPa
	X	−11.865 MPa	10.568 MPa
	Y	−28.328 MPa	43.635 MPa
	Z	−9.875 MPa	24.142 MPa
Contact Force			
	Total	0 N	3,158.63 N
	X	−921.015 N	445.997 N
	Y	−2,495.497 N	2,651.102 N
	Z	−1,707.459 N	1,173.31 N





3rd Principal



Displacement

Total

